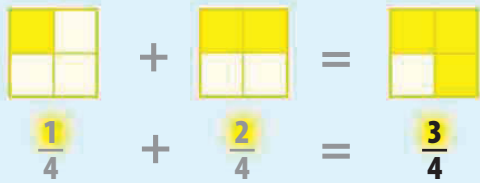


ADDING AND SUBTRACTING FRACTIONS

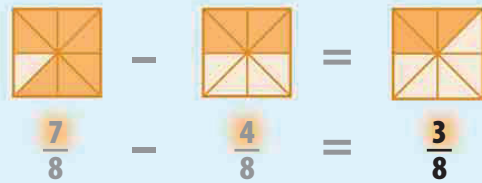
Just like whole numbers, it is possible to add and subtract fractions. How it is done depends on whether the denominators are the same or different.

Adding and subtracting fractions with the same denominator

To add or subtract fractions that have the same denominator, simply add or subtract their numerators to get the answer. The denominators stay the same.



To add fractions, add together only the numerators. The denominator in the result remains unchanged.



To subtract fractions, subtract the smaller numerator from the larger. The denominator in the result stays the same.

Adding fractions with different denominators

To add fractions that have different denominators, it is necessary to change one or both of the fractions so they have the same denominator. This involves finding a common denominator (see opposite).

multiply whole number by denominator
then add numerator

6 is a common
denominator of
both 3 and 6

$$4\frac{1}{3} + \frac{5}{6} \rightarrow \frac{4 \times 3 + 1}{3} \quad \frac{13}{3} + \frac{5}{6}$$

denominator stays same

First, turn any mixed fractions that are being added into improper fractions.

The two fractions have different denominators, so a common denominator is needed.

$\frac{5}{6}$ can now be added
to $\frac{26}{6}$ as both have
same denominator

remainder becomes
numerator of fraction

$$\frac{13}{3} = \frac{26}{6} + \frac{5}{6} \quad \frac{31}{6} \rightarrow 31 \div 6 = 5r1 = 5\frac{1}{6}$$

denominator goes into common denominator 2 times,
so multiply both numerator and denominator by 2

Convert the fractions into fractions with common denominators by multiplying.

If necessary, divide the numerator by the denominator to turn the improper fraction back into a mixed fraction.

Subtracting fractions with different denominators

To subtract fractions with different denominators, a common denominator must be found.

multiply whole number by denominator
then add numerator

4 is a common
denominator of
both 2 and 4

$$6\frac{1}{2} - \frac{3}{4} \rightarrow \frac{6 \times 2 + 1}{2} \quad \frac{13}{2} - \frac{3}{4}$$

denominator stays same

First, turn any mixed fractions in the equation into improper fractions by multiplying.

The two fractions have different denominators, so a common denominator is needed.

$\frac{3}{4}$ can be subtracted from
 $\frac{26}{4}$ because both have
same denominator

remainder becomes
numerator of fraction

$$\frac{13}{2} = \frac{26}{4} - \frac{3}{4} \quad \frac{23}{4} \rightarrow 23 \div 4 = 5r3 = 5\frac{3}{4}$$

denominator goes into common denominator 2 times,
so multiply both numerator and denominator by 2

Convert the fractions into fractions with common denominators by multiplying.

If necessary, divide the numerator by the denominator to turn the improper fraction back into a mixed fraction.